



Project Proposal

SECJ3623: Mobile Application Programming
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Project Title: Smart Recipe Finder
Group Name: NULL

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1. Project Background & Problem Statement

1.1 Background

With the increasing use of smartphones, many daily activities such as shopping, communication, and information sharing are now performed through mobile applications. Cooking is one of the daily activities where people frequently search for information online, especially food recipes. However, recipes are often scattered across websites, social media, notebooks, or messaging applications, making them difficult to organize and retrieve efficiently.

Small food business owners and cooking enthusiasts also face challenges in storing their recipes digitally while sharing them with others. A centralized mobile platform is needed to allow users to browse, save, and share recipes conveniently using their mobile devices.

1.2 Problem Statement

Currently, there is no simple mobile platform that allows users to browse recipes without login, while also supporting recipe sharing and management in one system. Many existing platforms require registration before access, which discourages casual users. Additionally, recipes stored manually or across different platforms may be lost, duplicated, or difficult to manage.

Therefore, a mobile application is required to provide an organized, user-friendly, and accessible recipe management system that supports different user roles and basic recipe-sharing functionality.

2. Objectives & Project Scope

2.1 Objectives

The objectives of this project are:

- To develop a mobile application that allows users to browse and search food recipes easily.
- To provide a platform for users to share and store recipes digitally.
- To enable the system owner to manage users and recipe content effectively.
- To support guest access for viewing recipes without requiring login.

2.2 Project Scope

The scope of this project includes:

- Development of a mobile application using **Flutter**.
- Backend services using **Firebase** for authentication and data storage.
- Support for three user roles: **Admin (Owner), User/Creator, and Guest**.
- Implementation of basic CRUD (Create, Read, Update, Delete) functions for recipes and users.

The project does **not include** payment processing, food delivery services, or advanced analytics features.

3. Target Users & User Needs

3.1 Target Users

The target users for this application are:

- **Admin (Owner):** Responsible for managing users and recipes.
- **User / Creator:** Users who browse, search, bookmark, and share recipes.
- **Guest:** Users who only want to view recipes without logging in.

3.2 Identified User Needs

- Admin needs a system to manage recipe content and user accounts.
- Users need a simple and fast way to browse, search, and save recipes.
- Users want to share their own recipes with others.
- Guests need the ability to view recipes without mandatory registration.
- All users require a simple and intuitive mobile interface.

4. Proposed Features

4.1 Core Features

A. Admin

- Manage users (view, activate/deactivate, or remove).
- Manage recipes (add, edit, delete, update).
- Oversee the overall app content and system maintenance.

B. User / Creator

- Register and log in to the app.
- Browse and search recipes.
- Bookmark favorite recipes.
- Add and share personal recipes (text, photo, or optional YouTube link).
- Manage personal profile information.

C. Guest

- Browse and view public recipes without logging in.
- Must register/login to bookmark or share recipes.

4.2 Technical Features

- Firebase Authentication for user login and access control.
- Cloud Firestore for storing recipe and user data.
- Firebase Storage for uploading recipe images.
- Role-based access control (Admin, User, Guest).

4.3 User Interface Features

- Clean and responsive mobile UI.
- Simple navigation structure (Home, Search, Profile).
- Recipe cards with images and descriptions.
- Easy-to-read recipe detail pages.
- Mobile-friendly layout suitable for all screen sizes.

5. Project Timeline & Milestones

5.1 Project Phases

Sprint 1 – Setup & Authentication

- Setup Flutter project & Firebase backend
- Implement login & registration (Admin & User)
- Create basic profile pages (Admin & User)
- Setup main navigation structure (Home, Login, Profile)

Sprint 2 – Recipe Management

- Admin: Manage recipes (Add, Edit, Delete, View)
- Admin: Add optional YouTube link for each recipe
- User/Creator: Add and share own recipes
User/Creator: View recipe list and recipe details

Sprint 3 – Recipe Interaction

- User/Creator: Search recipes by name or ingredient
- User/Creator: Bookmark favorite recipes (Add/Remove)
- User/Creator: View bookmarked recipes
- Guest: Browse public recipes without login

Sprint 4 – User Management & Maintenance

- Admin: Manage user accounts (View, Activate/Deactivate, Remove)
- User/Creator: Edit personal profile information

5.2 Development Milestones

Phase	Duration	Description	Status
Sprint 1	Week 1–4	Project setup, Firebase integration, authentication	Completed
Sprint 2	Week 5–8	Recipe management (CRUD)	Completed
Sprint 3	Week 9–11	Search, bookmark, and guest access	Completed
Sprint 4	Week 12–14	User management, testing, and final improvements	Completed

5.3 Development Team and Responsibilities

Name	Responsibilities
Mohammed Satippiddin Safid Bin Hamzah	Scrum Master
Mohamad Roslan Bin Saidi	Developer
Muhammad Asyraf Bin Ramlee	UI/UX Designer
Nurul Aidatul Aisyqin	Tester